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SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*
Assessment Tool Weightage: 40%

Ab-1.

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

	Q1 $L_3L_2L_1$	Q2 $L_3L_2L_1$	Q3 $L_3L_2L_1$	Q4 $L_3L_2L_1$	Q5 $L_3L_2L_1$	Q6 $L_3L_2L_1$	Q7 $L_3L_2L_1$	Q8 $L_3L_2L_1$	Q9 $L_3L_2L_1$	Q10 $L_3L_2L_1$
Understanding 45%	2.25	2	0		2	2.25	2.25			2.25
Explanation 35%	1.75	1.75	0		1			1.75		1
Clarity 20%	1	1	0		1			1		1
	5	4.75			4			5		4.25

23/50

CO-1 Assement tool - 1

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1. Given,

File size = 150 MB

Network Bandwidth = 50 Mbps.

Segment size = 1500 bytes.

Data link layer overhead = 40 bytes / frame.

Propagation delay is ignored.

Number of segments

The transport layer divides the file into segments of 1500 bytes.

File size :

150 MB = 150×10^6 = 150,000,000 bytes.

Number of segments :

$$\text{Segments} = \frac{150,000,000}{1500} = 100,000.$$

Since every segment becomes one frame

Number of frames = 1,00,000.

Each frame contains,

⇒ Data = 1500 bytes.

⇒ Header + trailer = 40 bytes

∴ Overhead 100,000 × 40 = 4,000,000 bytes.
= 4 MB.

Total Data Transmitted,

Original File = 150 MB

Protocol overhead = 4 MB.

∴ 150 + 4 = 154 MB.

Transmission time,

Convert total data into bits

154 MB ⇒ 154 × 8 = 1232 Mb.

Bandwidth = 50 Mbps.

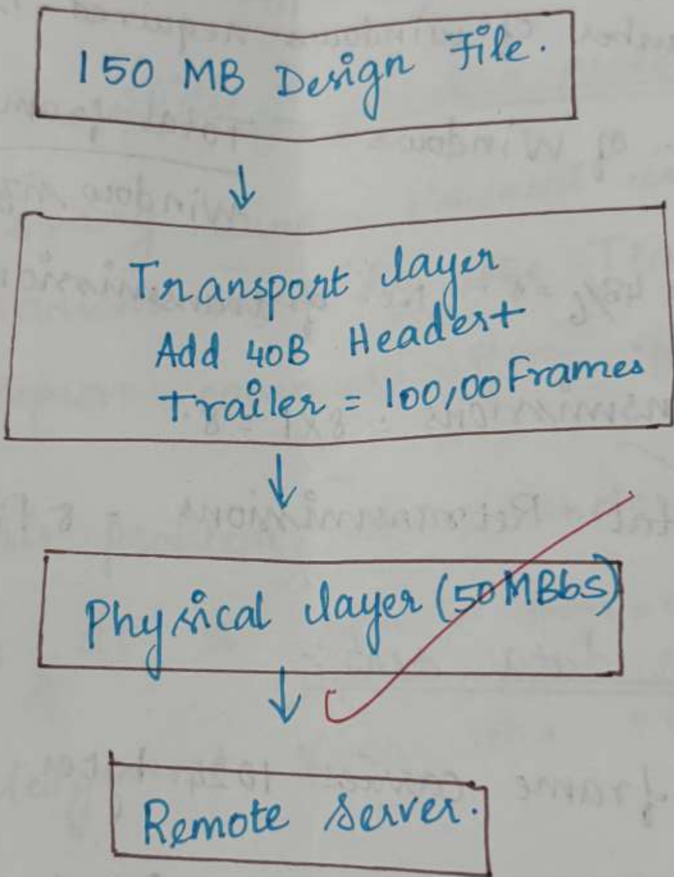
Transmission time = $\frac{1232}{50}$ = 24.64 seconds.

Explanation of the transport layer:

The transport layer is the fourth layer of the OSI Model and is responsible for providing reliable and end-to-end communication.

Explanation of the data link layer

The data link layer is the second layer of the OSI Model. It main responsibility is to provide reliable communication between two directly connected devices.



Conclusion: The transport layer & Data link layer work together to ensure accurate and reliable communication.

Final Answer:-

⇒ Window size = 6 frames.

⇒ Total frames = 48.

⇒ Transmission windows required = 8.

⇒ Retransmissions = 8 Frames

⇒ Total user data = 49,152 Bytes

Explanation of sliding window protocol:-

The sliding window Protocol is a flow control mechanism used by the Transport Layer to improve communication efficiency.

In this problem, the sender can transmit 6 frames at a time, before waiting for acknowledgement from the receiver.

How flow control improves communication efficiency,

Flow control is a technique that regulates the rate at which data is transmitted from the sender to the receiver.

2) Given Data:-

⇒ Window size = 6 frames.

⇒ User data per frame = 1024 bytes.

⇒ Total frames.

⇒ One frame is lost in every window.

Calculating the number of Transmission windows.

The number of windows required is calculated as

$$\text{No. of Windows} = \frac{\text{Total frames}}{\text{Window size}}$$

$= 48/6 = 8$. No. of transmission windows = 8.

Retransmissions $= 8 \times 1 = 8$.

∴ Total Retransmissions = 8 Frames.

Total user data sent:-

Each frame carries 1024 bytes.

Total user data transmitted:

$48 \times 1024 = 49,152$ bytes.

5) Session Layer,

Given data,

Total file size = 600 MB.

Checkpoint interval = 60MB

Failures occurs after = 390 MB.

Calculating Number of Checkpoints,

A checkpoint is created after every 60 MB.

Number of Checkpoints $= \frac{390}{60} = 6.5$

Number of Checkpoints $= 6$.

The last completed checkpoint is at,

$6 \times 60 = 360$ MB.

Calculate data to be Retransmitted,

Data sent after the last checkpoint,

$390 - 360 = 30$ MB.

Calculate data to be Retransmitted,

Data sent after the last checkpoint,

390 - 360 = 30 MB

Remaining data after failure:

600 - 390 = 210 MB.

Total data to be transmitted after recovery,

30 + 120 = 240 MB.

Final answer:-

Total checkpoints created = 6

Last checkpoint = 360 MB.

Data lost after last Checkpoint = 30 MB.

Remaining data = 210 MB.

Total data to be Retransmitted = 240 MB.

Importance of Synchronization in the session layer,

The session layer is the fifth layer of the OSI model and is responsible for establishing, managing & terminating communication sessions between two devices.

One of its most important functions is synchronization, inserting checkpoints.

Synchronization is essential for multimedia streaming, cloud storage, video conferencing.

7) Given data

Total data size = 3 GB

Effective throughput = 120 Mbps.

Request size = 3MB.

Converting GB into MB

1 GB = 1000 MB

∴ 3 GB = 3 × 1000 = 3000 MB.

Calculate number of Requests:

The Applications layer divides the file into requests of 3MB each,

Number of Requests $= \frac{3000}{3} = 1000$

$\therefore$ Total Requests Generated $= 1000$.

Calculate Transmission time:

Convert data into megabits (Mb).

Since, 1 Byte $= 8$ bits

$3000 \text{ MB} \times 8 = 24000 \text{ Mb}$.

Transmission time

$$= \frac{\text{Total Data}}{\text{Throughput}}$$

$$= \frac{24000}{120} = 200 \text{ seconds}$$

How the Application Layer provides reliable services,

The Application Layer is the topmost layer (Layer 7) of the OSI Model. It acts as the interface between the user and the network by providing services, such as FTP, HTTP, SMTP, DNS.

10) Given Data,

Original Medical Record = 240MB.

Compression = 30%.

Segment size = 1400 bytes.

Data link layer Overhead = 60 bytes/frame.

Calculate compressed file size,

The presentation layer compresses the file by 30%.

Compressed file size

= 240 × (100 - 30)

= 240 × 0.90

= 168 MB

Compressed file size = 168 MB.

Calculating number of segments.

convert compressed file into bytes

168 MB = 168,000,000 bytes

no. of segments

$$= \frac{168,000,000}{1400} = 120,000.$$

Calculate protocol overhead,

Each frame has 60 bytes of overhead.

Total overhead,

= 120,000 × 60 = 7,200,000 bytes

= 7.2 MB.

Final data transmitted

Final data

$168 + 7.2 = 175.2$ MB.

OSI layer work together:

The OSI (open systems Interconnection) model divides communication into seven layers. Each layer performs a specific task and work together to ensure secure, reliable and efficient data.

The application layer provides services to healthcare software and allows doctors and hospitals to access patients records over the IoT network.